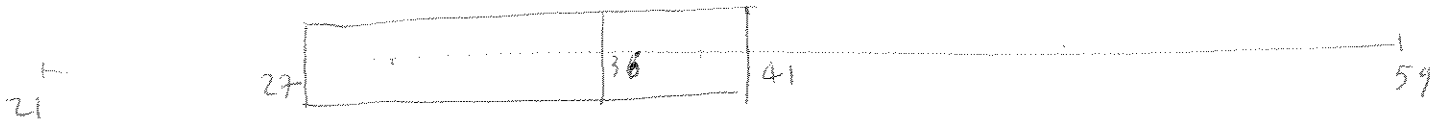


Meth 1040 Exam #2 Section 1

21 24 25 25 26 28 30 32 34 36 36 37 38 40 40 42 42 43 57 59
 $Q_1 = 27$ $Q_2 = 36$ $Q_3 = 41$



$$IQR = 41 - 27 = 14$$

$$38 \text{ years} \Rightarrow \frac{12}{20} = 0.6 \Rightarrow 60\text{th percentile}$$

$$25\text{th percentile } 0.25 \times 20 = 5 \quad 28 \text{ years old} = 2\text{th percentile}$$

$$P(\geq 1 \text{ head}) = 1 - P(\text{no heads}) = 1 - \left(\frac{1}{2}\right)^5 = 96.9\%$$

$$a) \frac{862 + 123 + 105}{1915} \approx 57\% \quad c) P(\text{Right} | \text{Male}) = \frac{862}{985} \approx 88\%$$

$$b) \frac{825}{1915} \approx 43\%$$

$$d) P(3 \text{ right handed} | \text{women}) = \frac{825}{1915} \times \frac{824}{1914} \times \frac{823}{1913} \approx 0.08 = 8\%$$

$$e) P(\text{Right hand}) = \frac{1687}{1915} \approx 88\%$$

$$P(\text{Right hand} | \text{Men}) = \frac{862}{985} \approx 88\%$$

} yes, independent

$$\textcircled{+} \frac{{}_9C_4}{{}_{16}C_4} = \frac{9 \cdot 8 \cdot 7 \cdot 6}{4 \cdot 3 \cdot 2 \cdot 1} \div \frac{16 \cdot 15 \cdot 14 \cdot 13}{4 \cdot 3 \cdot 2 \cdot 1} \approx 0.069 = 6.9\%$$

$$\textcircled{2} a) {}_{35}P_3 = 35 \cdot 34 \cdot 33 = 39270$$

$$\textcircled{6} \frac{7!}{2!3!2!} = 210$$

$$b) \frac{{}_{10}P_3}{{}_{35}P_3} = \frac{10 \cdot 9 \cdot 8}{35 \cdot 34 \cdot 33} \approx 0.018 = 1.8\%$$

$$c) {}_{35}C_3 = \frac{35 \cdot 34 \cdot 33}{3 \cdot 2 \cdot 1} = 6545$$

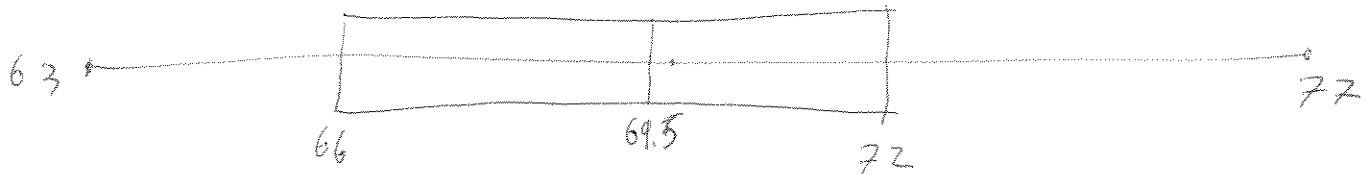
$$\textcircled{7} a) {}_{20}C_4 \times {}_5C_2 + {}_{20}C_5 \times {}_5C_1 + {}_{20}C_6 \times {}_5C_0 = \frac{20 \cdot 19 \cdot 18 \cdot 17}{4 \cdot 3 \cdot 2 \cdot 1} \times \frac{5 \cdot 4}{2 \cdot 1} + \frac{20 \cdot 19 \cdot 18 \cdot 17 \cdot 16}{5 \cdot 4 \cdot 3 \cdot 2 \cdot 1} \times \frac{5}{1} + \frac{20 \cdot 19 \cdot 18 \cdot 17 \cdot 16 \cdot 15}{6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1} = 48450 + 77520 + 38760 = 164730$$

$$b) \frac{{}_{20}C_3 \times {}_5C_3}{{}_{25}C_6} = \frac{\frac{20 \cdot 19 \cdot 18}{3 \cdot 2 \cdot 1} \times \frac{5 \cdot 4 \cdot 3}{3 \cdot 2 \cdot 1}}{\frac{25 \cdot 24 \cdot 23 \cdot 22 \cdot 21 \cdot 20}{6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1}} \approx 0.064 = 6.4\%$$

Math 1040 Exam #2 Section 2

① 63 64 64 65 66 66 67 68 68 69 70 70 70 71 72 72 72 74 75 77
 $\wedge \quad \quad \quad \wedge \quad \quad \quad \wedge$
 $Q_1 = 66 \quad \quad \quad Q_2 = 69.5 \quad \quad \quad Q_3 = 72$

$$IQR = 72 - 66 = 6$$



$$71 \text{ inches} \Rightarrow \frac{13}{20} = .65 \Rightarrow 65\text{th percentile}$$

$$25\text{th percentile} \Rightarrow \frac{5}{20} \Rightarrow 66 \text{ inches}$$

$$\textcircled{2} P(\text{at least 1 six}) = 1 - P(\text{no sixes}) = 1 - \left(\frac{5}{6}\right)^5 \approx .598 = 59.8\%$$

$$\textcircled{3} a) \frac{862 + 123 + 105}{1915} \approx 57\%$$

$$b) \frac{825}{1915} \approx 43\%$$

$$c) P(\text{Right} | \text{Male}) = \frac{862}{985} \approx 88\% \quad d) P(\text{3 right hand} | \text{women}) = \frac{825}{1915} \times \frac{824}{1914} \times \frac{823}{1913} \approx 8\%$$

$$e) P(\text{Right}) = \frac{1687}{1915} = 88\% = P(\text{Right} | \text{Male}) \quad \text{Yes; independent.}$$

$$\textcircled{4} \frac{{}^8C_3}{{}^{16}C_3} = \frac{8 \cdot 7 \cdot 6}{3 \cdot 2 \cdot 1} \div \frac{16 \cdot 15 \cdot 14}{3 \cdot 2 \cdot 1} = 10\%$$

$$\textcircled{5} a) {}_{20}P_3 = 20 \cdot 19 \cdot 18 = 6840 \quad b) \frac{{}_{10}P_3}{{}_{20}P_3} = \frac{10 \cdot 9 \cdot 8}{20 \cdot 19 \cdot 18} \approx 0.105 = 10.5\%$$

$$c) {}_{20}C_3 = \frac{20 \cdot 19 \cdot 18}{3 \cdot 2 \cdot 1} = 1140$$

$$\textcircled{6} \frac{10!}{4! \cdot 2! \cdot 3! \cdot 1!} = \frac{10 \cdot 9 \cdot 8 \cdot 7 \cdot 6 \cdot 5}{2 \cdot 3 \cdot 2 \cdot 1} = 12600$$

$$\textcircled{7} a) {}_{20}C_4 \times {}_5C_2 + {}_{20}C_5 \times {}_5C_1 + {}_{20}C_6 \times {}_5C_0 = \frac{20 \cdot 19 \cdot 18 \cdot 17}{4 \cdot 3 \cdot 2 \cdot 1} \times \frac{5 \cdot 4}{2 \cdot 1} + \frac{20 \cdot 19 \cdot 18 \cdot 17 \cdot 16}{5 \cdot 4 \cdot 3 \cdot 2 \cdot 1} \times \frac{5}{1}$$

$$+ \frac{20 \cdot 19 \cdot 18 \cdot 17 \cdot 16 \cdot 15}{6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1} = 48450 + 77520 + 38760 = 164730$$

$$b) \frac{{}_{20}C_4 \times {}_5C_2}{{}_{25}C_6} = \frac{48450}{\frac{25 \times 24 \times 23 \times 22 \times 21 \times 20}{6 \times 5 \times 4 \times 3 \times 2 \times 1}} \approx 0.27 = 27\%$$